DTC P045B:00 [SKYACTIV-D 1.5]

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DTC P045B:00	Low pressure EGR valve control: high duty malfunction
DETECTION CONDITION	• The LP-EGR control valve drive current exceeds the specification or the LP-EGR control valve drive circuit temperature is **175 °C {347 °F} or more** for a countinuous **2 s** when the following condition is met. **MONITORING CONDITIONS** — PCM internal processor malfunction DTC P0606:00 not detected. **Diagnostic support note** • This is an intermittent monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Low pressure EGR valve connector or terminals malfunction • PCM connector or terminals malfunction • Low pressure EGR valve position sensor (built-into low pressure EGR valve) malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED REPAIR INFORMATION AVAILABILITY** • Verify related Service Bulletins and/or on-line repair information availability. • Is any related repair information available?	Yes	Perform repair or diagnosis according to the available repair information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT LOW PRESSURE EGR VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the low pressure EGR valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 6.
		No	Go to the next step.
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 6.
		No	Go to the next step.
5	**INSPECT LOW PRESSURE EGR VALVE POSITION SENSOR** • Reconnect all disconnected connectors. • Inspect the low pressure EGR valve position sensor. (See LP-EGR CONTROL VALVE POSITION SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction**?**	Yes	Replace the low pressure EGR valve, then go to the next step. (See LP-EGR CONTROL VALVE POSITION SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
6	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
7	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.